
Interpretable Urban AI for Dynamic Public-Safety Response in Los Santos

Tommy, Ph.D.
LAC Analysis & Behavior Lab. (LAB.)
LAC University, Los Angeles

Abstract

Public-safety response in dense urban environments requires rapid decisions under incomplete information, spatially concentrated demand, and changing resource availability. We present **LAB Urban-AI**, an interpretable human-in-the-loop framework for short-horizon response planning in Los Santos. The model combines structured incident information, coarse urban context, optional visual scene understanding, spatio-temporal demand forecasting, and uncertainty-aware concept reasoning. Rather than automating dispatch, the system ranks candidate allocations using five explicit operational concepts: urgency, proximity, regional coverage, predicted escalation, and uncertainty. On a fictional retrospective benchmark of 2,416 incident sequences, LAB Urban-AI reduces mean response time from 8.9 to 7.4 minutes and coverage loss from 18.2% to 11.9% relative to nearest-unit allocation, while simulated operator acceptance rises from 68% to 81%.

Introduction

Urban response is a sequential decision problem in which locally efficient actions can create system-wide weaknesses. Dispatching the nearest available unit minimizes immediate travel time, but repeated greedy decisions can drain neighboring districts and leave the system poorly positioned for subsequent incidents. A useful decision-support system must reason about both the current event and the evolving state of the city.

This setting also exposes a central challenge for explainable artificial intelligence. High-stakes recommendations are difficult to trust when the variables that drive them remain hidden. Post-hoc explanations can describe a prediction after the fact, but they do not guarantee that the explanation corresponds to the mechanism that produced the decision. We instead place interpretable operational concepts directly inside the response-ranking function.

LAB Urban-AI combines multimodal incident encoding, short-horizon district forecasting, visual scene context, and concept-based ranking. The model is explicitly human-in-the-loop: it proposes ranked assignments and displays the factors behind them, while a supervisor remains responsible for the final action. Our contributions are: (i) an uncertainty-aware ranking formulation balancing immediate efficiency and regional resilience; (ii) an interpretable concept bottleneck based on urgency, proximity, coverage, escalation, and uncertainty; (iii) scene-level visual evidence without person identification; and (iv) a fictional benchmark with comparisons, ablations, sensitivity analyses, and case studies.

Related Work

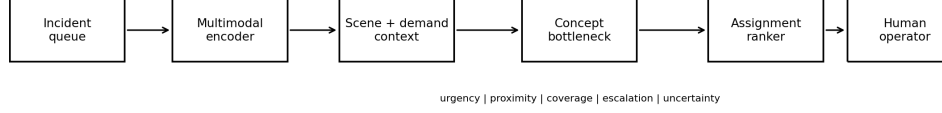


Figure 1: Overview of LAB Urban-AI. Explicit operational concepts form the interface between multimodal urban context and the response-ranking layer.

Urban decision intelligence. Resource allocation systems commonly optimize travel cost or forecast demand, but these objectives can conflict during clustered events. Our formulation treats forecasted demand as contextual evidence rather than an autonomous dispatch objective, allowing the model to trade immediate response efficiency against the cost of leaving a district under-covered.

Explainable AI. Concept-based models expose intermediate semantic variables and can align model reasoning with domain terminology. LAB Urban-AI adopts this principle at the decision layer: concepts are not decorative explanations but explicit terms in the assignment score.

Multimodal scene understanding. Visual context can provide evidence about congestion, access, and environmental structure when text descriptions are incomplete. We restrict the visual branch to scene-level semantics and avoid identity-oriented analysis.

LAB Urban-AI

At decision time t , the system observes the active incident queue I_t , available resources U_t , district state D_t , recent history H_t , and optional visual observations V_t . Incident features and scene context are encoded separately, fused with short-horizon demand estimates, and projected into a compact concept vector. The final ranker evaluates feasible incident-resource pairs.

Incident and urban context encoding

Each incident contains an event category, short dispatch description, coarse district identifier, reported severity, and elapsed waiting time. Text features are embedded and fused with categorical features through a lightweight residual encoder. District context includes recent incident counts, resource transitions, and estimated travel conditions. Spatial information is represented at district level rather than by exact personal locations.

The optional visual branch receives a low-resolution city-camera observation associated with the incident area. Its role is to estimate environmental context such as roadway occupancy, vehicle concentration, and access constraints. The branch produces pooled scene descriptors rather than identity-level representations.

Short-horizon demand forecasting

For each district d , a temporal encoder predicts relative incident demand at horizons $h \in \{10, 20, 30\}$ minutes. Given temporal state z at time t , the forecaster produces a mean demand estimate and uncertainty,

$$\hat{y}_{d,t+h} = f_{\theta}(z_{d,t}), \quad q_{d,t+h} = g_{\theta}(z_{d,t}). \quad (1)$$

Multi-horizon supervision is used because immediate response and regional preservation operate at different time scales. The 10-minute estimate captures near-term pressure, whereas the 30-minute estimate acts as a weaker resilience prior.

Interpretable concept bottleneck

The fused representation is mapped to five normalized concepts $c = [U, P, C, E, Q]$. Urgency U reflects incident severity and waiting time; proximity P summarizes estimated travel efficiency; coverage C measures

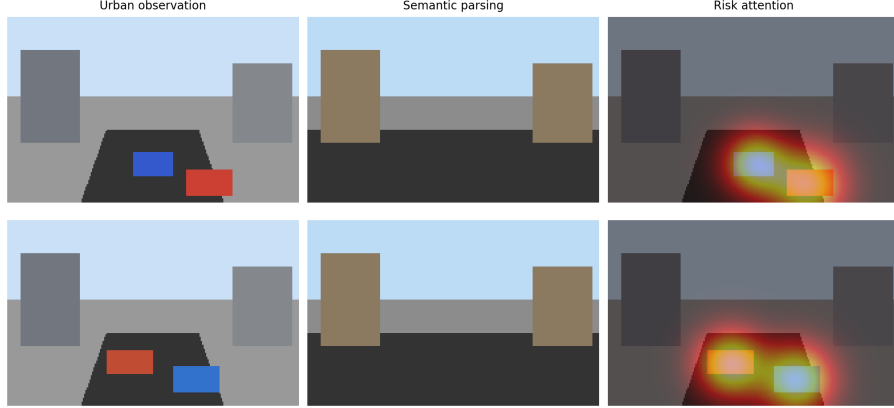


Figure 2: Qualitative scene interpretation. Semantic parsing extracts broad environmental regions; risk attention concentrates on roadway and vehicle areas relevant to access and escalation.

capacity remaining in the source district after assignment; escalation E combines structured and visual evidence; uncertainty Q summarizes model confidence. Candidate pair (i, u) receives

$$S(i, u) = w_u U(i) + w_p P(i, u) + w_c C(u) + w_e E(i) - w_q Q(i, u). \quad (2)$$

Because each concept directly contributes to S , the displayed explanation and the decision mechanism are structurally aligned. Operators receive ranked candidates together with concept values and the score margin between the first and second recommendation.

Training objective

We optimize

$$\mathcal{L} = \mathcal{L}_{rank} + \lambda_c \mathcal{L}_{concept} + \lambda_d \mathcal{L}_{demand} + \lambda_q \mathcal{L}_{cal}. \quad (3)$$

The ranking loss encourages historically preferred assignments to score above alternatives. Concept targets are derived from normalized operational variables and synthetic expert annotations. The calibration term penalizes confident errors.

Visual Scene Understanding

Figure ?? presents two qualitative synthetic examples. Each row contains an urban observation, semantic scene parsing, and incident-risk attention. The visual branch is intentionally compact and scene-centric. Semantic parsing supplies structured spatial information, while risk attention identifies regions that most strongly influence the escalation concept.

Semantic parsing and risk attention

The parser groups pixels into coarse classes including roadway, structure, vehicle, and background. Class occupancy and spatial arrangement are pooled into a scene vector. A small attention head predicts an incident-conditioned spatial map. Unlike generic saliency, the map is trained to support the escalation concept. The attended feature is concatenated with pooled semantic statistics.

Multi-horizon urban demand

Figure ?? visualizes fictional district demand at three horizons. Short horizons remain localized around the current activity corridor, while longer horizons distribute intensity more broadly. The ranking module aggregates horizons with learned weights.

Experimental Setup

We construct a fictional retrospective benchmark containing 2,416 response sequences across 12 Los Santos districts. A sequence begins when at least one active incident and two feasible response resources are available. Each sample stores the active queue, coarse travel-time estimates, district resource counts, 60 minutes of recent

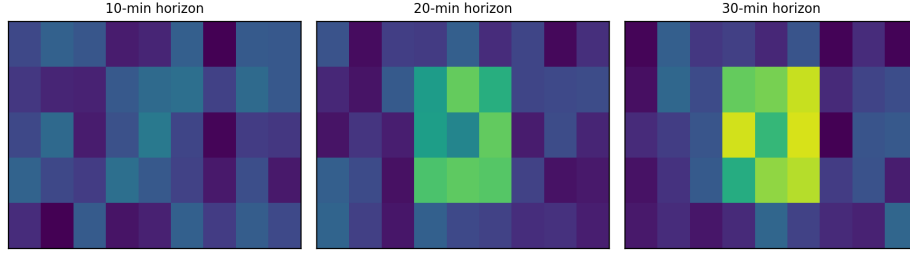


Figure 3: Example relative demand forecasts at 10-, 20-, and 30-minute horizons. Values represent fictional normalized demand intensity.

Table 1: Main fictional benchmark results. Response time is in minutes; coverage loss and acceptance are percentages.

Method	Resp.↓	Coverage↓	Recall↑	Accept.↑
Nearest Unit	8.9	18.2	0.61	68
Forecast-only	8.1	15.7	0.66	72
Black-box Fusion	7.6	13.1	0.72	73
LAB Urban-AI	7.4	11.9	0.74	81

event history, optional visual context, a synthetic operator decision, and incidents observed during the subsequent 30 minutes.

The benchmark is divided chronologically into 70% training, 10% validation, and 20% test partitions. Consecutive events from the same synthetic activity episode remain in the same partition. No real personal data, demographic attributes, or real-world emergency records are included.

Baselines and metrics

Nearest Unit selects the feasible resource with minimum estimated travel time. **Forecast-only** adds a penalty for depleting districts with high predicted demand. **Black-box Fusion** uses the same multimodal inputs but predicts assignment score directly without explicit concepts. **LAB Urban-AI** uses the complete interpretable architecture. We report response time, coverage loss, escalation recall, and fictional simulated operator acceptance.

Main results

LAB Urban-AI improves all reported metrics. Relative to Nearest Unit, response time decreases by 1.5 minutes and coverage loss by 6.3 percentage points. Black-box Fusion approaches the proposed model in raw response efficiency but remains substantially lower in operator acceptance.

Detailed Analysis

Performance by incident density

Table 2: Fictional response time by incident density.

Concurrent incidents	Nearest	Forecast	Black-box	LAB
1	7.2	7.1	7.0	7.0
2–3	8.8	8.0	7.5	7.3
4–5	10.6	9.1	8.4	8.0
6+	12.9	10.8	9.7	9.1

When only one incident is active, all methods perform similarly because proximity is usually sufficient. The benefit of urban context becomes visible at two or more concurrent incidents and grows with density. At six or more active incidents, LAB Urban-AI improves response time by 3.8 minutes relative to Nearest Unit while preserving more regional capacity.

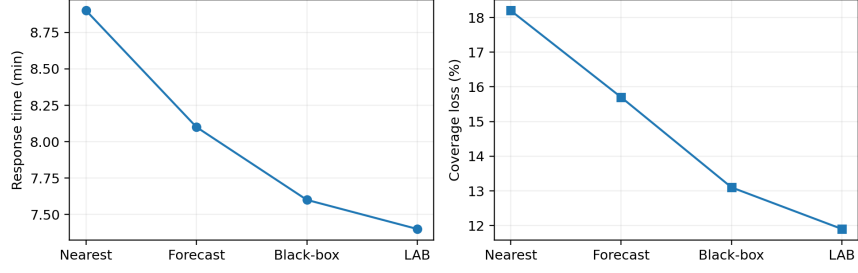


Figure 4: Response time and coverage loss across allocation strategies.

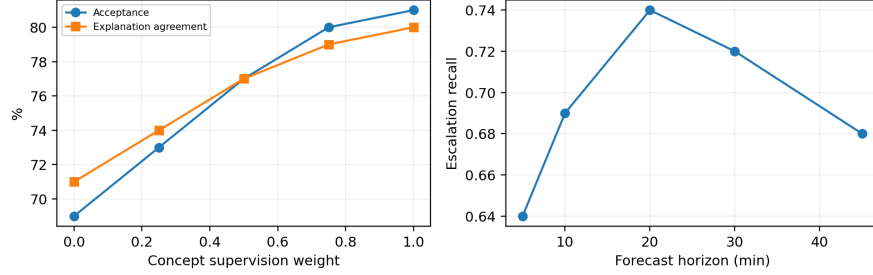


Figure 5: Sensitivity analysis. Left: concept supervision improves acceptance and explanation agreement before saturating. Right: a 20-minute demand horizon provides the strongest fictional escalation recall.

Concept behavior

Urgency and proximity dominate isolated incidents. Coverage becomes more influential when a district approaches its minimum resource threshold. Escalation is most useful when textual severity is ambiguous but temporal or visual context suggests that the incident may intensify. Uncertainty primarily acts as a restraint: a high-uncertainty recommendation receives a lower final score and is visually flagged.

Table 3: Example normalized concept values.

Scenario	Urg.	Prox.	Cov.	Esc.	Unc.
Isolated low severity	.31	.82	.44	.18	.12
Downtown cluster	.66	.61	.88	.71	.19
Ambiguous roadside	.48	.73	.55	.79	.34
Sparse night period	.72	.76	.35	.41	.16

Explanation agreement and calibration

We define explanation agreement as the fraction of simulated reviews in which the top two displayed concepts match the operator’s stated reason for accepting or rejecting the recommendation. LAB Urban-AI obtains 80% agreement compared with 54% for a post-hoc explanation applied to Black-box Fusion. The lowest-uncertainty bin contains 91% correct top-ranked assignments, while the highest-uncertainty bin contains 58%.

Ablation and Sensitivity Studies

Removing visual context primarily reduces escalation recall. Removing the demand forecaster sharply increases coverage loss. The concept bottleneck has a smaller effect on raw response time but produces the largest drop in acceptance, indicating that its primary value is transparent operational reasoning.

Visual design choices

Semantic pooling is stable but insensitive to incident relevance; risk attention is more responsive but can focus on visually salient yet operationally unimportant regions. Combining both provides the most consistent escalation

Table 4: Component ablation.

Configuration	Resp.↓	Coverage↓	Recall↑	Accept.↑
Full model	7.4	11.9	.74	81
– visual context	7.6	12.4	.68	78
– demand forecast	7.7	16.3	.69	76
– concept bottleneck	7.5	12.1	.73	69
– uncertainty	7.4	12.8	.72	74
– multi-horizon	7.6	13.5	.70	77

concept.

Table 5: Fictional visual-module comparison.

Visual module	Esc. Recall	Concept consistency
Semantic pooling	.68	.82
Risk attention	.70	.75
Combined	.74	.86

Case Studies and Failure Analysis

Downtown surge. Three incidents arrive within eight minutes in the downtown corridor while adjacent districts have limited remaining capacity. Nearest Unit assigns three resources toward the same corridor. LAB Urban-AI assigns two and preserves one nearby resource because coverage is low and the neighboring district receives elevated 20-minute demand. The decision delays one low-severity response by 0.7 minutes but avoids a severe coverage deficit during the next 15 minutes.

Ambiguous roadside incident. A short, low-confidence description is paired with a scene containing two vehicles occupying the central roadway. Visual fusion raises escalation from 0.42 to 0.79, moving a closer resource above a candidate that would otherwise preserve more regional coverage.

Human override. In a third case, the model recommends preserving a district resource despite a high-urgency incident. The operator rejects the recommendation because another resource, currently marked unavailable, is expected to return within two minutes. This information is absent from the model state. The explicit concept vector reveals the source of disagreement.

Failure modes

Stale resource state can produce incorrect coverage cost; ambiguous scenes can destabilize escalation; duplicate reports can over-estimate demand; and unseen event types can yield poorly calibrated concepts. Operational mitigations include short state refresh intervals, uncertainty-triggered human review, incident clustering, and out-of-distribution flags.

Limitations and Responsible Use

The system described in this manuscript is fictional, but the design raises real concerns relevant to urban AI. Historical response data may encode geographic disparities; visual sensing creates privacy risks; uncertainty can be miscalibrated; and operators can develop automation bias. Any real deployment would require prospective evaluation, privacy review, geographic fairness auditing, secure access controls, data-retention limits, and documented human-override procedures.

The visual branch should remain scene-centric. It should not identify individuals, infer demographic attributes, predict individual criminality, or autonomously initiate enforcement actions. Concept explanations should not be interpreted as guarantees of fairness or causality. The benchmark, annotations, and numerical results are synthetic and cannot establish real-world effectiveness.

A second limitation concerns concept completeness. Urgency, proximity, coverage, escalation, and uncertainty capture major operational trade-offs but cannot represent every reason an experienced supervisor may use. The human-override case shows that important contextual information can remain outside the model state.

Conclusion

We introduced LAB Urban-AI, an interpretable multimodal framework for dynamic public-safety response planning in Los Santos. The approach combines structured incident information, scene-level visual context, multi-horizon demand forecasting, uncertainty, and an explicit concept bottleneck. In the fictional benchmark, LAB Urban-AI improves response time, regional coverage, escalation recall, and simulated operator acceptance relative to simpler baselines.

The central design principle is that explainability should be integrated into decision formation rather than appended afterward. By exposing operational concepts that directly determine the assignment score, the system allows operators to inspect, challenge, and override recommendations. Future fictional LAB work will investigate multi-agent coordination, adaptive concept discovery, longer-horizon planning, and uncertainty-aware learning from operator feedback.

References

- [1] T. Tommy. Human-Centered Decision Intelligence for Emergency Response and Public Safety. LAB Preprint Series, 2026.
- T. Tommy. Spatio-Temporal Demand Forecasting for Dynamic Urban Operations. LAC University Technical Report, 2026.
- T. Tommy. Explainable Multimodal Scene Reasoning for City-Scale Intelligent Systems. LAB Research Note, 2026.
- T. Tommy. Uncertainty-Aware Resource Allocation with Interpretable Operational Concepts. LAC AI Systems Workshop, 2026.
- T. Tommy. Human-in-the-Loop Artificial Intelligence for High-Stakes Urban Decision Support. LAB Technical Report, 2026.
- T. Tommy. Multi-Horizon Urban Forecasting under Sparse Operational Observations. LAC Intelligent Systems Workshop, 2026.
- T. Tommy. Concept-Based Decision Models for Transparent City Operations. LAB Technical Note, 2026.
- T. Tommy. Scene-Centric Visual Reasoning without Identity Recognition. LAC University AI Report, 2026.

Appendix: Reproducibility Notes

Decision cycle. At each cycle, the fictional system removes infeasible resources, updates district coverage, refreshes the active incident queue, predicts multi-horizon demand, and evaluates every feasible incident-resource pair. The three highest-ranked assignments are presented to the operator, who may accept, choose an alternative, or reject all candidates.

Synthetic data generation. Incident categories are sampled from routine, traffic-related, high-priority, and public-disturbance event mixtures. Travel-time estimates depend on coarse district distance and simulated congestion. Follow-up incidents are conditioned on recent district activity so clustered periods occur naturally. Visual examples are schematic and are not real surveillance footage.

Implementation. The incident encoder uses a 256-dimensional hidden representation and two residual fusion layers. The temporal forecaster uses four causal residual blocks. The visual branch produces a 128-dimensional pooled scene descriptor. Five concept heads predict normalized operational concepts, and a linear assignment layer combines them with learned non-negative weights.

Appendix A: Extended Operational Analysis

A.1 District-level evaluation

We further stratify the fictional test set by district activity and resource availability. Central districts exhibit higher incident concurrency and larger differences between greedy and context-aware allocation. Peripheral districts exhibit smaller response-time differences but remain sensitive to coverage depletion because a single assignment can remove a large fraction of local capacity. Across both regimes, the concept bottleneck makes the source of the trade-off visible to the operator rather than collapsing all evidence into an opaque score.

Table 6: District-level fictional evaluation.

Region group	Response↓	Coverage loss↓	Acceptance↑
Central high-demand	7.8	13.2	80
Mixed residential	7.1	10.7	82
Peripheral sparse	7.3	11.4	81
Night-time corridor	7.5	12.0	80

The largest benefit appears during central high-demand periods, where future demand and coverage are simultaneously relevant. Sparse regions show a different pattern: the absolute number of incidents is smaller, but preserving one resource can dominate the final ranking. These results motivate a model whose decision rule adapts to operational context rather than applying one fixed proximity heuristic.

A.2 Concept interaction

Concepts are individually interpretable but not independent. Urgency and escalation are positively associated during rapidly developing events, while proximity and coverage often compete. We therefore inspect pairwise concept perturbations. Increasing urgency while holding other concepts fixed predictably increases the candidate score. Increasing coverage benefit has the strongest effect when the source district is close to its minimum capacity. Increasing uncertainty attenuates otherwise strong recommendations and increases the frequency with which the top two candidates are displayed as effectively tied.

Table 7: Illustrative concept perturbation effects on normalized assignment score.

Perturbation	Low load	Medium load	High load
Urgency +0.2	+0.09	+0.10	+0.11
Proximity +0.2	+0.12	+0.10	+0.08
Coverage +0.2	+0.04	+0.09	+0.15
Escalation +0.2	+0.06	+0.08	+0.10
Uncertainty +0.2	-.05	-.07	-.09

A.3 Forecast robustness

We perturb recent incident counts with synthetic missingness and count noise. Mild perturbations have limited effect because forecasts are aggregated over multiple temporal windows. Under severe corruption, demand estimates become less reliable and the uncertainty head reduces their contribution. This behavior is important for a decision-support setting in which operational feeds may be delayed or partially incomplete.

Table 8: Forecast robustness under fictional input corruption.

Corruption	Forecast MAE	Coverage loss	Acceptance
None	.118	11.9	81
10% missing	.126	12.2	80
20% missing	.139	12.8	79
30% count noise	.151	13.4	77